# THE MEDICAL DATA HISTORY PROJECT

Based Patient and Admissions Data Analysis

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# CONTENT:

- **Objective:** Analyze patient and hospital data using SQL queries.
- **Database Used:** project_medical_data_history
- **Tables Included:**
    - Patients
    - Admissions
    - Doctors
    - Province names
- **Goal:** Retrieve insights on patients' demographics, diagnoses, and hospital admissions.

MySQL Workbench

project_medical_data_history

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

- project_medical_data_histo
  - Tables
    - admissions
    - doctors
    - patients
    - province_names
  - Views
  - Stored Procedures
  - Functions

Administration Schemas

Information

**No object selected**

SQL File 1*

Limit to 1000 rows

```sql
Use project_medical_data_history;
select * from admissions;
select * from doctors;
select * from patients;
select * from province_names;

-- 1. Show first name, last name, and gender of patients who's gender is 'M'
select first_name, last_name, gender from patients where gender = "M";

-- 2. Show first name and last name of patients who does not have allergies.
select first_name, last_name from patients where allergies is null;

-- 3. Show first name of patients that start with the letter 'C';
select first_name from patients where first_name like 'C%';

-- 4. Show first name and last name of patients that weight within the range of 100 to 120 (inclusive)
select first_name, last_name from patients where weight between 100 and 120;

-- 5. Update the patients table for the allergies column. If the patient allergies is null then replace it with "NKA"
update patients set allergies = "NKA" where allergies is null;

select first_name, last_name,coalesce(allergies, 'NKA') as allergies from patients;

-- 6. Show first name and last name concatenated into one column to show their full name.
select *,concat(first_name,' ',last_name) as full_name from patients;

-- 7. Show first name, last name, and the full province name of each patient.
select a.first_name, a.last_name, b.province_name, b.province_id from patients a join province_names b on a.province_id = b.province_id;

```

Output

Object Info Session

SQL Editor closed

MySQL Workbench

project_medical_data_history

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Navigator

SCHEMAS

Filter objects

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Object Info Session

SQL File 1*

Limit to 1000 rows

```sql
-- 8. Show how many patients have a birth_date with 2010 as the birth year.
select COUNT(*) as total_patients_2010 from patients where year(birth_date) = 2010;

-- 9. Show the first_name, last_name, and height of the patient with the greatest height.
select first_name, last_name, height from patients order by height desc;

-- 10. Show all columns for patients who have one of the following patient_ids: 1,45,534,879,1000
select * from patients where patient_id in (1,45,534,879,1000);

-- 11. Show the total number of admissions
select count(*) from admissions;

-- 12. Show all the columns from admissions where the patient was admitted and discharged on the same day.
select * from admissions where admission_date = discharge_date;

-- 13. Show the total number of admissions for patient_id 579.
select count(patient_id) from admissions where patient_id = 579;

-- 14. Based on the cities that our patients live in, show unique cities that are in province_id 'NS'?
select distinct city from patients where province_id = "NS";

-- 15. Write a query to find the first_name, last name and birth date of patients who have height more than 160 and weight more than 70
select first_name, last_name , birth_date from patients where height > 160 and weight > 70;

-- 16. Show unique birth years from patients and order them by ascending.
select distinct year(birth_date) as birth_year from patients order by birth_year;
```

Output

SQL Editor closed

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SQL File 1*

```sql
-- 17. Show unique first names from the patients table which only occurs once in the list.
-- For example, if two or more people are named 'John' in the first_name column then don't include their name in the output list.
-- If only 1 person is named 'Leo' then include them in the output. Tip: HAVING clause was added to SQL because the WHERE keyword cannot be used with aggregate functions.
select first_name from patients group by first_name having count(first_name) = 1;

-- 18. Show patient_id and first_name from patients where their first_name start and ends with 's' and is at least 6 characters long.
select patient_id, first_name from patients where first_name like 'S%S' and length(first_name) >= 6;

-- 19. Show patient_id, first_name, last_name from patients whos diagnosis is 'Dementia'. Primary diagnosis is stored in the admissions table.
select a.patient_id, a.first_name, a.last_name from patients a join admissions b on a.patient_id = b.patient_id where b.diagnosis = 'dementia';

-- 20. Display every patient's first_name. Order the list by the length of each name and then by alphbetically.
select first_name from patients order by length(first_name),first_name;

-- 21. Show the total amount of male patients and the total amount of female patients in the patients table. Display the two results in the same row.
select sum(case when gender = 'M' then 1 else 0 end) as total_male, sum(case when gender = 'F' then 1 else 0 end) as total_female from patients;

-- 22. Show patient_id, diagnosis from admissions. Find patients admitted multiple times for the same diagnosis.
select patient_id, diagnosis, count(*) as admission_count from admissions group by patient_id,diagnosis having count(*) > 1;

-- 23. Show the city and the total number of patients in the city. Order from most to least patients and then by city name ascending.
select city, count(patient_id) as Total_patients from patients group by city order by Total_patients desc, city asc;

-- 24. Show first name, last name and role of every person that is either patient or doctor. The roles are either "Patient" or "Doctor"
select first_name, last_name, 'patients' as role from patients union select first_name, last_name, 'Doctors' as role from doctors;

-- 25. Show all allergies ordered by popularity. Remove NULL values from query.
select allergies, count(*) as total_patients from patients where allergies is not null group by allergies order by total_patients desc;
```

Output

SQL Editor closed

MySQL Workbench

project_medical_data_history ×

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Object Info Session

SQL File 1* ×

Limit to 1000 rows

```sql

-- 26. Show all patient's first_name, last_name, and birth_date who were born in the 1970s decade. Sort the list starting from the earliest birth_date.
select first_name, last_name, birth_date from patients where year(birth_date) between 1970 and 1979 order by birth_date asc;

-- 27. We want to display each patient's full name in a single column. Their last_name in all upper letters must appear first, then first_name in all lower case letters.
-- Separate the last_name and first_name with a comma. Order the list by the first_name in decending order    EX: SMITH,jane
select concat(upper(last_name),',',lower(first_name)) as Full_Name from patients order by first_name desc;

-- 28. Show the province_id(s), sum of height; where the total sum of its patient's height is greater than or equal to 7,000.
select province_id, sum(height) as Total_Height from patients group by province_id having Total_Height >= 7000;

-- 29. Show the difference between the largest weight and smallest weight for patients with the last name 'Maroni'
select max(weight) - min(weight) as weight_difference from patients where last_name = 'Maroni';

-- 30. Show all of the days of the month (1-31) and how many admission_dates occurred on that day. Sort by the day with most admissions to least admissions.
select day(admission_date) as day_of_month, count(*) as total_admissions from admissions group by day(admission_date) order by total_admissions desc;

-- 31. Show all of the patients grouped into weight groups. Show the total amount of patients in each weight group. Order the list by the weight group decending.
-- e.g. if they weight 100 to 109 they are placed in the 100 weight group, 110-119 = 110 weight group, etc.
select (weight / 10) * 10 as weight_group, COUNT(*) as total_patients from patients group by weight_group order by weight_group desc;

-- 32. Show patient_id, weight, height, is Obese from the patients table. Display isObese as a boolean 0 or 1. Obese is defined as weight(kg)/(height(m).
-- Weight is in units kg. Height is in units cm
select patient_id, weight, height, case when (weight / POWER(height / 100.0, 2) >= 30) then 1 else 0 end as Obese from patients;

-- 33. Show patient_id, first_name, last_name, and attending doctor's specialty. Show only the patients who has a diagnosis as 'Epilepsy' and the doctor's first name is 'Lisa'.
-- Check patients, admissions, and doctors tables for required information.
select a.patient_id, a.first_name, a.last_name,c.specialty from patients a join admissions b on a.patient_id = b.patient_id
 join doctors c on b.attending_doctor_id = c.doctor_id where b.diagnosis = 'Epilepsy' and c.first_name = 'Lisa';
```

Output

Action Output

Query Completed

# CORE SQL QUERIES & OPERATIONS

- **Data Filtering:**
  - Patients with gender = ‘M’
  - Patients without allergies
- **Data Transformation:**
  - Concatenating names
  - Calculating BMI (obesity indicator)
- **Data Cleaning:**
  - Replacing NULL allergies with “NKA”

# ANALYTICAL SQL QUERIES

- **Grouping & Aggregation:**
  - Count of patients per city
  - Total admissions
  - Weight group distribution
- **Statistical Insights:**
  - Tallest patient
  - Patients born in the 1970s
  - Height/Weight analysis by province

# COMPLEX QUERY OPERATIONS

- **Join Operations:**
- Linking patients with doctors and admissions
- **Examples:**
  - Patients diagnosed with *Epilepsy* and attended by Dr. Lisa
  - Unique cities by province
  - Repeated admissions for same diagnosis

# PROJECT OUTCOME & INSIGHTS

- **Results:**
  - Extracted patient trends and hospital statistics effectively
  - Applied SQL functions, joins, and conditional logic
- **Learnings:**
  - Improved understanding of relational databases
  - Gained proficiency in writing optimized SQL queries
- **Future Scope:**
  - Integrate visualization (Power BI / Tableau)
  - Automate report generation